

A Graph-based Error Correction Model Using Lie-algebraic Cohomology and Its Application on Global Consistent Indoor Scene Reconstruction

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ABSTRACT

We present a novel, effective method for global indoor scene reconstruction problems by geometric topology. Based on point cloud pairwise registration methods (e.g ICP) or IMU, we focus on the problem of accumulated error for the composition of transformations along any loops. The major technical contribution of this paper is a linear method for the graph optimization, using only solving a Poisson equation. We demonstrate the consistency of our method from Hodge-Helmholtz decomposition theorem and experiments on multiple RGBD datasets of indoor scene. The experimental results also demonstrate that our global registration method runs quickly and provides accurate reconstructions.

Index Terms: Human-centered computing—Visualization—Visualization techniques—3D Scene Reconstruction; Human-centered computing—Visualization—Visualization design and evaluation methods—Modeling and simulation

1 INTRODUCTION

Recently, indoor three-dimensional reconstruction has a strong application in the field of virtual reality (VR), which have been widely used in education, telecommuting, tourism, and many other industries. The goal of this theme is to build a three-dimensional model of the indoor scenes based on the data collected by various sensors, mathematical tools it uses include multiview geometry, probability statistics, and optimization theory, e.t.c. When collecting large amounts of multi-view data, it is inevitable that drifts will appear, they are caused by the error of pose estimation, and the error always increases as the number of poses increases. To improve the global consistency, the fundamental concept of *viewing graph* has been developed. The viewing graph encapsulates the poses treated as nodes and the relative transformations between them as edges. *Global consistency* is defined as concatenating transformations along each loop in the graph. It should return the identity under the assumption of an ideal noise-free setting. Based on the viewing graph, many iterative reconstruction methods have been proposed [2, 5, 8]. While the existing methods cannot achieve global consistency with high precision, only obtain rough approximated solutions. Our goal is to propose a novel algorithm to tackle the loop closing problem and greatly improve the global consistency.

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2 OUR METHOD

In this work, we approach the global consistency problem from a fundamentally different perspective: *the viewing graph is treated not as a solely 1-skeleton graph, but a higher dimensional Cech complex, namely it has much richer topological information, encoded by the Cech cohomology, which is the key to solve the loop closing problem.* The pipeline of our method is shown in Fig. 1.

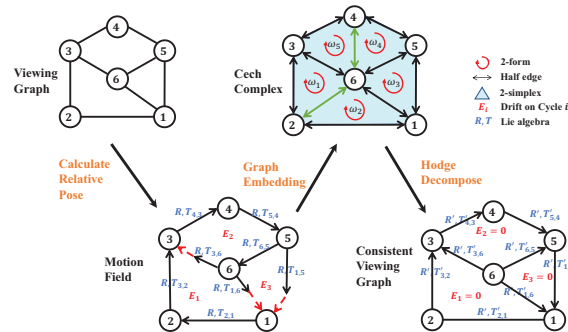


Figure 1: Hodge Algorithms Pipeline. a) Construct viewing graph by pairwise SIFT, b) calculate Lie algebra of rigid transform from ICP, c) triangularization by graph embedding, 4) solve curl free field. Finally, we realize consistency.

In more detail, our algorithm is based on the following four steps:

- Construct viewing graph** The viewing graph $G = (V_G, E_G)$ is constructed as usual, each node represents a point cloud, the corresponding camera frame is called a pose. The point clouds are roughly registered using ICP or NDT algorithm. Each edge represents two intersecting point clouds, and is associated with the transformation between the two poses associated with the end nodes.
- Graph Embedding** The topological structure of the Cech complex is complicated, involving high dimensional simplexes. For Hodge decomposition, only 2-skeleton is needed. Therefore, we can embed the view graph on a surface, and use the surface to approximate the 2-skeleton of the Cech complex and perform the Hodge decomposition. This method preserves the algebraic property of the viewing graph [4], ensures the loop closing condition and improves the efficiency.
- Rigid Transformation 1-Form** All the rigid motions form a Lie group, the tangent space at the identity form the Lie algebra. For each oriented edge of the complex, we associate a rigid motion with it, which can be estimated by the iterative closest point (ICP) algorithm. Then we further represent the rigid motion as a vector in the Lie algebra. Therefore we can treat the rigid motions defined on all the edges as a Lie-algebraic simplicial 1-form on the viewing graph then use the Lie-algebraic simplicial cohomology to rephrase

the loop close problem: *the integration of the Lie-algebraic 1-form is zero along any closed loop (1-chain), this implies the 1-form is exact, namely it is the gradient of some function defined on the viewing graph [4].*

d. Helmholtz-Hodge decomposition The well-known Helmholtz-Hodge decomposition theorem [3, 6, 7] claims that any differential 1-form on a Riemannian manifold can be decomposed into three orthogonal components, an exact form (curl free), a co-exact form (divergence free) and a harmonic form (both curl free and divergence free). Therefore, we can extract the exact component from the initial Lie-algebraic 1-form.

3 EXPERIMENTAL EVALUATIONS

We compare the reconstruction results obtained by different algorithms, including the “ground truth” (GT), the BundleFusion (BF) [1], and our Hodge decomposition algorithm with one iteration (Hodge 1st) and three iterations (Hodge 3rd). The comparison results are summarized in Table 1 and Table 2.

Table 1: On the viewing graph under Euclidean metric, the average negative exponential power of the point-2-plane distance of ground truth, BundleFusion and Hodge method of iteration one and three time. The score’s upper bound is 1, the larger the better.

	GT	BF	Hodge 1st	Hodge 3rd
chess	0.829	0.827	0.827	0.827
fire	0.869	0.868	0.870	0.869
redkitchen	0.863	0.865	0.865	0.865
stairs	0.821	0.818	0.822	0.822
living2	0.803	0.883	0.921	0.918
office2	0.783	nan	0.931	0.937

Table 2: On the viewing graph under IoU metric, the average score of ground truth, BundleFusion and Hodge method of iteration one and three time. The score’s upper bound is 1, the larger the better.

	GT	BF	Hodge 1st	Hodge 3rd
chess	0.837	0.835	0.836	0.838
fire	0.870	0.868	0.870	0.870
redkitchen	0.862	0.864	0.864	0.865
stairs	0.818	0.814	0.820	0.820
living2	0.805	0.878	0.919	0.915
office2	0.781	nan	0.930	0.935

Fig. 2 shows the comparison among ground truth, BundleFusion [1] and our algorithm on the ICL-NUIM dataset. By examining the reconstruction details, one can see that our method achieves the best quality and has the least drifting effects.

4 CONCLUSION

This work introduces a novel algorithm to tackle the challenging loop closing problem in global point cloud registration and its application on global consistent indoor scene reconstruction. The algorithm generalizes the viewing graph to a Cech complex with richer topological structure, and treat the relative transformations defined on the edges as a Lie-algebraic simplicial 1-form on the Cech complex, then use Helmholtz-Hodge decomposition to extract the exact component of the 1-form, which gives the relative transformations among the frames. Furthermore, the 2-skeleton of the Cech complex can be approximated by an embedding surface of the viewing graph.

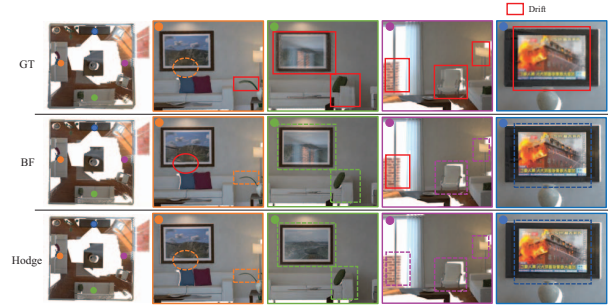


Figure 2: Comparison among the “ground truth”, the BundleFusion [1] and our algorithm on the Living Room 2 scene in the ICL-NUIM dataset. Top row: the reconstruction (point cloud) by the “ground true” and the four local details (safron: painting and plants, green: painting and sofa, purple: Balcony, armchair and lamp, blue: TV). Middle row: the reconstruction by the BundleFusion algorithm. Bottom row: the reconstruction by our method. The solid red box indicates poor reconstruction due to the drifting effect, and the dotted box indicates the better results. It can be observed that our approach has the least drifts and highest global consistency.

Experimental results demonstrate the simplicity, efficiency and accuracy of the algorithm. It outperforms the conventional methods and achieves higher global consistency.

However, this algorithm assumes the relative transformations on each edge in the viewing graph is in the neighborhood of the identity in the Lie group of the rigid motion, hence it can be represented using a vector in the Lie algebra. If the poses are sparse, and the relative transformations are far away for the identity, the Lie-algebraic representations are inaccurate. The proposed method will encounter difficulties. In the future, we consider to combine data collected by alignment sensors, such as IMU, to overcome this shortcoming.

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REFERENCES

- [1] A. Dai, M. Nießner, M. Zollhöfer, S. Izadi, and C. Theobalt. Bundlefusion: Real-time globally consistent 3d reconstruction using on-the-fly surface reintegration. *ACM Transactions on Graphics (ToG)*, 36(4):1, 2017.
- [2] D. Crandall, A. Owens, N. Snavely, and D. Huttenlocher. Discrete-continuous optimization for large-scale structure from motion. In *CVPR 2011*, pp. 3001–3008, 2011.
- [3] H. Bhatia, G. Norgard, V. Pascucci, and P. Bremer. The helmholtz-hodge decomposition—a survey. *IEEE Transactions on Visualization and Computer Graphics*, 19(8):1386–1404, 2013.
- [4] J. Lee. *Smooth manifolds*. Springer, 2013.
- [5] J. Lim, J. Frahm, and M. Pollefeys. Online environment mapping. In *CVPR 2011*, pp. 3489–3496. IEEE, Colorado Springs, CO, USA, June 2011.
- [6] L. Lui, W. Zeng, S. T. Yau, and X. Gu. Shape analysis of planar multiply-connected objects using conformal welding. *IEEE transactions on pattern analysis and machine intelligence*, 36(7):1384–1401, 2014.
- [7] M. Desbrun, E. Kanso, and Y. Tong. Discrete differential forms for computational modeling. In *Discrete differential geometry*, pp. 287–324. Springer, 2008.
- [8] N. Snavely, S. Seitz, and R. Szeliski. Skeletal graphs for efficient structure from motion. In *2008 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–8, 2008.